Tyler Mitts

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Education

University of Missouri – Columbia Expected May 2027
B.S. in Computer Science, Minor in Mathematics and IT (GPA: 3.6/4.0)

Research Experience

University of Missouri, Division of Plant Science and Technology Sept 2025 – Present
Undergraduate Research Assistant – Machine Learning for Plant Genomics & Phenomics

DNA to Display: Genotype-to-Image Generative Modeling *PyTorch, Latent Diffusion, SLURM*

- Built a genotype-to-image system (LiteVAE + cross-attention latent diffusion U-Net) generating maize root cross-sections and seed images from genome-wide SNPs (43,788-gene MexMAGIC; 90,790-SNP WiDiv)
- Benchmarked on a 1,051-line synthetic population with 42 designed QTL and leakage-free splits: on unseen lines, generated stele diameter tracked true genetic value at $r = 0.53$ (GBLUP 0.62)
- Designed memorization tests (denoising with own vs. least-related genotype): genotype use now carries to unseen lines (+1.24% vs. +1.63% on training lines), where early models showed none
- Designed a windowed haplotype encoder: red seeds for 90% of unseen red-seed genotypes vs. 15% with PCA; seed redness $r = 0.85$ vs. 0.69 for GBLUP; planted stele gene on held-out lines $r = 0.69$ (PCA 0.58)
- Traced seed-dependent broken generations to the noise schedule's nonzero terminal SNR; retraining the autoencoder and matching the starting noise raised valid generated roots from 55% to 79%

The Cortical Cell Pattern – tylermitts.com/root_pattern *YOLOv8, OpenCV, SciPy, statsmodels*

- Built a 7-stage pipeline with three fine-tuned YOLOv8 models (root detection, cell instance segmentation, cortex masking) and human-in-the-loop QC, measuring 1.44M cells in 1,708 roots across 4 species
- Discovered a conserved radial cortical cell-size pattern (spline descriptors validated by functional PCA); built a derivative-based cell-file detector (Savitzky–Golay smoothing, 1-D k-means)
- Tested the pattern with ANOVA/Tukey HSD and mixed-effects models, with exploratory TDA (Vietoris–Rips)

Industry Experience

Mitts Technology LLC – ApplyApply.org *(Founder & Full-Stack Developer)* Nov 2025 – Present

- Built a subscription SaaS job-application platform (React, AWS serverless) with LLM document generation via AWS Bedrock, Stripe payments, rate limiting, and caching

Projects

Vision-Guided CNC Automation *PyTorch, YOLOv10-seg, OpenCV, Streamlit, GRBL*

- Trained a custom YOLOv10 segmentation model to detect poker-chip inlay offsets in real time; built camera-to-machine calibration and serial G-code correction for fully automated drilling

Microscopy Cell & Nuclei Segmentation *OpenCV, scikit-image, NumPy*

- Separated and counted touching cells in breast-cancer and malaria microscopy (k-means, watershed); isolated nuclei in cancer and kidney-tumor images with multi-scale LoG blob detection

Transformer-Based 3D Reconstruction *PyTorch, VGGT, Open3D, OpenCV*

- Reconstructed 3D point clouds from image sequences with the VGGT transformer (depth, pose, and points in one pass); evaluated pose error against ground truth and voxelized with confidence filtering

WhereToStargaze.us – 1st Place, TigerHacks Hackathon 2025 *React, FastAPI, Redis*

- Ranked stargazing sites from light pollution and cloud cover data; Redis caching cut load times by 40%

Leadership & Teaching

- Taught a summer computer-vision study group for peers; mentored a first-year student's TDA research project

Technical Skills

Languages: Python, C++, C, Rust, Java, JavaScript, SQL
ML & Data: PyTorch, diffusion models, VAEs, YOLO, scikit-learn, OpenCV, TensorFlow, NumPy, pandas
Genomics & Stats: SNP encoding, PCA, kinship/GBLUP, QTL benchmarking, mixed models, fPCA, TDA
Infrastructure: SLURM/HPC, CUDA, Git, AWS (Lambda, S3, Bedrock), FastAPI, React, Redis, PostgreSQL